Neuro-Symbolic Language Modeling with Automaton-augmented Retrieval

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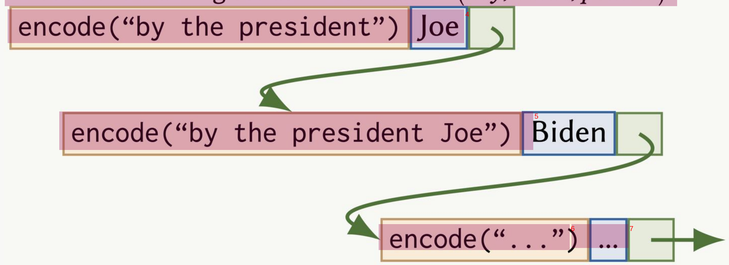
https://github.com/neulab/retomaton https://github.com/neulab/knn-transformers

Fairseq implement. Huggingface implement. PDF

Key Idea #1: Pointers Between Examples

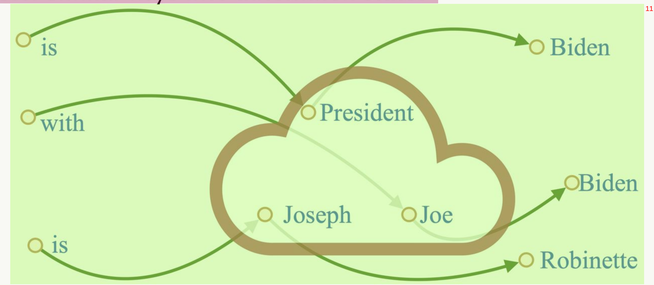
Encode the training set as linked lists of (*key, value, pointer*):



This extends *k*NN-LM (Khandelwal et al., '2020) by adding *pointers* between consecutive datastore entries.

Key Idea #2: Clustering Similar Keys

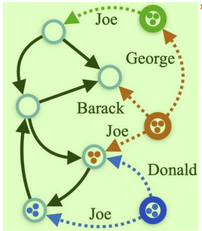
Cluster similar keys into automaton states:



RETOMATON

Test context: The president is ____

Automaton

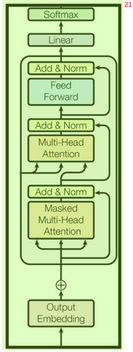


Nodes: clusters of training examples, encoded by the LM

Edges: pointers between consecutive examples, shared in cluster

Weights: $-\|h_t, \text{key}\|_2$

Trained LM



$$\lambda P_{auto} + (1-\lambda) P_{LM}$$

$$P_{auto}(w \mid h_t, states_t) \propto \sum_{s \in states_t} \sum_{(key,val) \in s, val=w} exp(-\|h_t, \text{key}\|_2)$$

In-domain Datastore

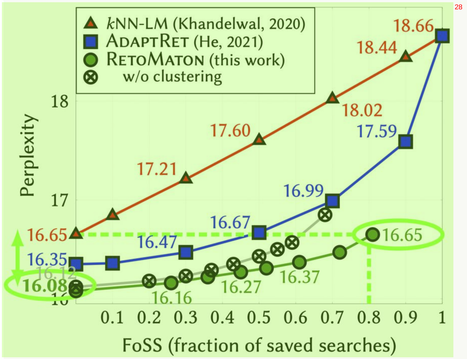


Figure: Experiments on WIKITEXT-103, where the datastore is created from the same training set that the base LM was trained on.

Domain Adaptation

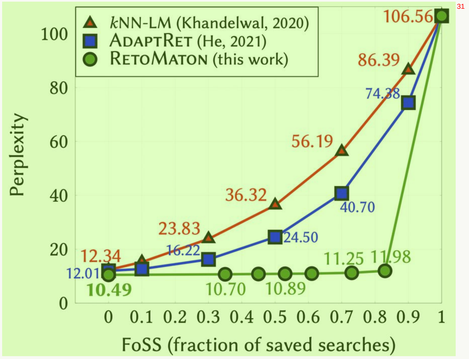


Figure: Domain adaptation experiments: the model was trained on News Crawl, and the datastore is constructed from Law-MT.

Improving Fine-Tuning

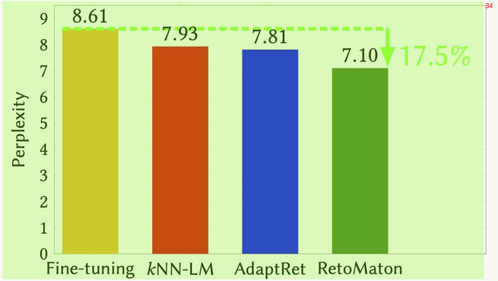


Figure: When constructing RETOMATON on top of a fine-tuned model, RETOMATON reduces perplexity by 17.5%.

Sample

Figure: A random sample of the automaton constructed from the training set of WIKITEXT-103